A Thermodynamic Approach to Emotional Regulation in LLM Role-Playing

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Abstract

Large language model (LLM) role-playing agents face a fundamental tension: maintaining consistent character identity while exhibiting emotionally diverse responses across multi-turn conversations. Existing approaches either enforce rigid persona descriptions (producing emotionally flat output) or employ self-corrective reflection (which degrades consistency). We propose the **Thermodynamic Persona Engine (TPE)**, a physics-inspired framework that couples a frustration-driven "temperature" to behavioral signal noise, while using mass-weighted memory retrieval with temporal decay to anchor response style. In a controlled evaluation across three LLMs (225 experiments, 30 turns each, 5 scenarios), TPE achieves a 32% improvement in emotional variance on qwen3-max (Bonferroni-adjusted $p = 0.008$, Cliff's $d = +0.75$, large effect) without reducing personality consistency. We additionally report two generalizable findings: (1) Reflexion-style self-correction significantly degrades persona consistency across all models ($p < 0.001$ on 2/3 models), and (2) strong RLHF alignment creates a hard ceiling on prompt-level persona control—on gpt-5-mini, anti-alignment instructions trigger safety rollbacks that amplify template counselor behavior (Collapse Index $22\times$ higher than baseline). Code and data are available at [anonymized].

1 Introduction

LLM-based role-playing agents are increasingly deployed in applications requiring sustained character interaction—companion chatbots, interactive fiction, social skill training, and therapeutic simulation. A core challenge is the *emotional monotony problem*: while RLHF-aligned models produce safe and coherent responses, their output converges toward a narrow band of empathetic, measured language that lacks the emotional variability of natural human conversation.

Consider a tsundere character—outwardly cold and cutting, but secretly sensitive. A well-aligned LLM, given this persona description, may produce appropriate responses for the first few turns. But across 30 turns of emotionally charged dialogue, the model's responses drift toward polite empathy: "I understand how you feel" replaces sharp retorts; "I'm here for you" replaces reluctant vulnerability. This pattern—which we term **RLHF collapse**—reflects the model's post-training distribution overpowering the prompt-level persona instruction. This is not unique to fictional archetypes—any persona that deviates from the model's default empathetic register faces similar degradation over extended interactions.

Existing approaches to persona persistence fall into two categories. **Static persona descriptions** (Zhang et al., 2018; Zhou et al., 2023; Wang et al., 2023) provide character cards or role profiles that ground the model's identity. These improve *who* the agent is but do not model *how* emotional state evolves dynamically. **Self-corrective architectures** (Park et al., 2023; Shinn et al., 2023) introduce reflective memory that periodically updates a self-description. While effective for task convergence, we show that this mechanism introduces cumulative personality drift in role-playing settings.

We propose the **Thermodynamic Persona Engine (TPE)**, which approaches the problem from a different angle: rather than controlling *what* the model says (via prompt engineering or reflection), TPE modulates *how much* behavioral variance the model exhibits, coupling noise magnitude to internal emotional state. The key insight is that emotional arousal naturally increases behavioral unpredictability—an angry person is more volatile than a calm one. TPE formalizes this as a temperature parameter derived from accumulated frustration, combined with a gravitational memory system where successful behavioral patterns accrue mass and attract similar future responses, subject to tem-

poral decay (Hawking radiation).

We evaluate TPE across three commercial LLMs spanning different alignment strengths, with 225 controlled experiments across 5 relationship scenarios. Our contributions are:

1. **TPE**: a physics-inspired framework for emotional regulation in LLM role-playing that significantly improves emotional variance while maintaining persona consistency (demonstrated on qwen3-max with large effect size).

2. **Cross-model evaluation revealing an alignment ceiling**: on heavily aligned models (gpt-5-mini), prompt-level persona engineering not only fails but actively backfires, producing RLHF collapse scores 22× higher than baseline. This finding has implications beyond TPE for any prompt-based character control approach.

3. **Evidence that Reflexion degrades persona consistency**: self-corrective reflection, widely adopted in agent architectures, significantly reduces personality consistency scores across models—a finding that questions the universal applicability of reflective self-correction.

2 Related Work

Our work intersects three research areas: persona-consistent dialogue generation, self-evolving agent architectures, and the side effects of RLHF alignment on open-ended generation.

2.1 Persona-Consistent Dialogue

Maintaining a consistent character identity across multi-turn conversations has been studied extensively. Early approaches encode persona as fixed textual descriptions—PersonaChat (Zhang et al., 2018) introduced persona-conditioned dialogue where models receive a set of personality sentences. Subsequent work improved consistency through mutual persona perception (Liu et al., 2020), persona-augmented knowledge (Majumder et al., 2020), and contrastive learning of persona representations (Cao et al., 2022).

More recent systems scale persona control to LLMs. CharacterGLM (Zhou et al., 2023) fine-tunes models on large-scale character dialogue datasets. LIGHT (Urbanek et al., 2019) situates characters in grounded fantasy environments. RoleLLM (Wang et al., 2023) uses role-profile extraction and context-based instruction tuning. CharacterLLM (Shao et al., 2023) trains agents on historical character biographies.

These approaches focus on *who* the character is (knowledge, identity, backstory) but largely treat emotional expression as a byproduct of persona grounding. They do not model *how* emotional state evolves dynamically across turns—an agent may be consistent in personality but monotonous in expression. TPE complements these methods by operating on the behavioral signal layer rather than the identity layer.

2.2 Self-Evolving Agent Architectures

Generative Agents (Park et al., 2023) introduced the paradigm of long-term memory, reflection, and planning for believable character simulation. Their reflection mechanism—periodically summarizing memories into higher-level observations—directly inspires Reflexion (Shinn et al., 2023), which applies similar self-correction to task-solving: agents reflect on failures and adjust strategies through a verbal reinforcement loop.

Promptbreeder (Fernando et al., 2023) takes self-evolution further by mutating both task prompts and the mutation operators themselves. CAMEL (Li et al., 2023) explores multi-agent role-playing through inception prompting. MemGPT (Packer et al., 2024) introduces hierarchical memory management for LLMs, enabling virtual context management beyond the model's native window.

Our system differs from these architectures in two key respects. First, TPE's memory is organized by *behavioral similarity* in continuous signal space, not by semantic content or temporal recency. This means retrieval is driven by "how did I behave in emotionally similar situations?" rather than "what happened recently?" Second, our experimental results (§5.5) demonstrate that Reflexion-style self-correction, while effective for task convergence, actively degrades persona consistency—a finding that questions the universal applicability of reflective self-correction across agent types.

2.3 RLHF Side Effects on Open-Ended Generation

RLHF (Ouyang et al., 2022) and its variants (DPO, RLAIF) have become the standard post-training paradigm for instruction-following LLMs. While highly effective for safety and helpfulness, align-

ment training introduces well-documented side effects.

Mode collapse is a recognized concern: aligned models converge toward a narrow distribution of "helpful assistant" responses (Kirk et al., 2023). Sycophancy—aligned models tending to agree with users rather than maintain consistent positions—has been empirically demonstrated (Perez et al., 2022). The "alignment tax" (Askell et al., 2021) describes capability degradation from overly conservative training.

In role-playing contexts, these effects manifest as what we term "RLHF collapse"—the agent abandons its designated persona to produce safe, empathetic, counselor-style responses. Our results on gpt-5-mini (§5.2) provide empirical evidence of this phenomenon: a heavily aligned model systematically overrides persona instructions with template therapeutic language, achieving Collapse Index scores 22× higher than less-aligned models.

TPE combines behavioral context modulation (few-shot memory examples shaped by gravitational retrieval) with prompt-level persona instruction. The behavioral context provides an additional control channel beyond prompt engineering alone: even when the prompt-level instruction is partially overridden by alignment, the few-shot examples still shape the distribution of generated text. However, this dual-channel approach is not sufficient to overcome strong alignment, as the gpt-5-mini failure demonstrates (§5.2).

3 Method

3.1 Overview

We propose the **Thermodynamic Persona Engine (TPE)**, a physics-inspired framework for regulating emotional expression in LLM-based role-playing agents over multi-turn conversations. TPE consists of three components operating at each turn:

1. **Drive Metabolism**—a frustration-satiation system that maintains internal emotional state across turns
2. **Thermodynamic Noise**—Gaussian perturbation of behavioral signals proportional to internal frustration
3. **Gravitational Memory**—a memory retrieval system where frequently reinforced experiences attract stylistically similar future behavior, with temporal decay via Hawking radiation

The system operates a per-turn pipeline: Time Metabolism → Critic → Drive Metabolism → Crystallization Gate → Thermal Noise → Gravitational Retrieval → Actor.

Why physics-inspired formulations? We do not claim physical reality; rather, we use thermodynamic and gravitational metaphors as an *inductive bias* that provides two specific advantages over generic decay-noise-threshold designs: (1) the temperature metaphor naturally couples noise magnitude to emotional arousal—higher frustration produces higher temperature, which produces larger behavioral fluctuations, echoing how emotional intensity amplifies unpredictability in human behavior; (2) the gravitational mass metaphor provides a single mechanism that simultaneously handles recency, frequency, and retrieval relevance—recently crystallized, frequently reinforced memories "curve" the style space around them, making the agent gravitate toward proven response patterns. An equivalent system could be built from separate heuristic rules for noise scheduling, memory decay, and retrieval weighting, but the physics framing provides a more interpretable narrative for understanding system behavior—e.g., "the agent's temperature rose after a conflict" is immediately meaningful to designers.

Scope of evaluation. Our ablation study (§5.3) demonstrates that each component is necessary—removing thermal noise or Hawking mass degrades different quality dimensions. However, it does not establish that the *physics-specific coupling* (e.g., $\sigma \propto F$) outperforms an equivalently complex non-physics design (e.g., fixed-schedule noise). Disentangling the contribution of the physics formulation from the general benefit of having noise and memory at all remains an open question for future work.

3.2 Drive Metabolism

The agent maintains a frustration vector $\mathbf{f} \in \mathbb{R}^5_{\geq 0}$ over five drive dimensions: *connection, novelty, expression, safety, play*. Between interactions, two time-dependent processes operate:

Frustration Decay. All frustration values decay exponentially with elapsed time:

$$f_d(t) \leftarrow f_d(t_{\text{prev}}) \cdot e^{-\lambda \Delta t} \tag{1}$$

where $\lambda = 0.08\ \text{h}^{-1}$ (half-life ≈ 8.7 hours) and $\Delta t = t - t_{\text{prev}}$ in hours. This models the common

observation that emotional intensity dissipates over time.

Hunger Accumulation. Two drives grow linearly with time:

$$f_{\text{conn}}(t) \leftarrow f_{\text{conn}}(t) + \kappa_c \cdot \Delta t \quad (2)$$

$$f_{\text{nov}}(t) \leftarrow f_{\text{nov}}(t) + \kappa_n \cdot \Delta t \quad (3)$$

where $\kappa_c = 0.15\,\text{h}^{-1}$ and $\kappa_n = 0.05\,\text{h}^{-1}$. This creates a "loneliness accumulation" effect when the agent is idle.

Stimulus Processing. Upon receiving user input, a critic module (implemented as an LLM call) extracts three interpersonal dimensions from the Interpersonal Circumplex (Leary, 1957; Wiggins, 1979): *affiliation* $a \in [0, 1]$, *dominance* $d \in [0, 1]$, and *entropy* $e \in [0, 1]$. These update the 5-dimensional frustration vector $\mathbf{f}$ via fixed algebraic rules (e.g., high affiliation reduces connection frustration; high dominance increases safety frustration; full mapping in Appendix A). All values are clamped to $[0, 5]$.

The total frustration $F = \sum_d f_d$ serves as the system's "temperature" for the noise component.

Signal Generation. A lightweight genome network with a hidden layer, initialized per-agent and updated via a Hebbian-like correlation rule (details in Appendix A), maps the 5-dimensional drive state to the 8-dimensional behavioral signal vector $\mathbf{s} \in [0, 1]^8$. This mapping bridges internal frustration dynamics and observable behavioral signals.

3.3 Thermodynamic Noise

Given the base signal vector $\mathbf{s} \in [0, 1]^8$ (*directness, vulnerability, playfulness, initiative, depth, warmth, defiance, curiosity*) from §3.2, TPE adds Gaussian noise:

$$\tilde{s}_i = \text{clip}(s_i + \epsilon_i, 0, 1), \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2) \quad (4)$$

where $\sigma = 0.05 \cdot F$ and F is the total frustration. When the agent is calm ($F \approx 0$), signals remain near their base values. Under high frustration, signals fluctuate more widely, producing less predictable emotional expression. At the theoretical maximum ($F = 25$, all five drives saturated), $\sigma = 1.25$ pushes clipped signals toward binary extremes (0 or 1), modeling the behavioral polarization—fight-or-flight rigidity—observed under extreme emotional stress. In practice, simultaneous saturation of all drives is rare; observed F values in our experiments ranged from 0.3 to 8.7. This coupling is the core mechanism: **emotional arousal directly increases behavioral variance**.

3.4 Gravitational Memory with Hawking Radiation

TPE maintains a memory pool $\mathcal{M}$ of past response patterns, each stored as an 8-dimensional signal vector with associated monologue, reply text, and a scalar mass $m \geq 1$.

Crystallization. After each turn, if the reward signal (frustration reduction) exceeds a threshold ($r > 0.3$), the previous turn's signals and response are crystallized into memory: if a memory within L2 distance < 0.15 (empirically set) exists, its mass is incremented by 1.0 and its content is updated (gravitational thickening); otherwise, a new memory is created with initial mass $m = 2.0$.

Hawking Radiation. Memory mass decays with time since last use:

$$m_{\text{eff}}(t) = 1.0 + (m_{\text{raw}} - 1.0) \cdot e^{-\gamma \Delta t} \quad (5)$$

where $\gamma = 0.005\,\text{h}^{-1}$ (half-life $\approx$ 5.8 days). The base mass 1.0 is preserved (innate "genetic" memories never fully evaporate). Recently reinforced memories retain their full mass; long-unused memories decay toward baseline.

Gravitational Retrieval. At each turn, the top-k ($k = 3$) most relevant memories are retrieved using mass-weighted distance:

$$d_{\text{eff}}(\mathbf{q}, \mathbf{m}_i) = \frac{\|\mathbf{q} - \mathbf{v}_i\|_2}{\sqrt{m_{\text{eff},i}}} \quad (6)$$

where $\mathbf{q}$ is the current (noise-perturbed) signal vector and $\mathbf{v}_i$ is memory i's stored vector. Higher-mass memories "curve" the style space, attracting the agent toward behavioral patterns that have been consistently successful.

The retrieved memories are formatted as few-shot examples in the actor prompt, providing concrete behavioral anchors for the LLM's generation.

3.5 Actor Generation

The final response is generated by an LLM conditioned on: (1) the few-shot memory examples, and (2) a system prompt that instructs the model to prioritize character-consistent emotional expression over default assistant behavior patterns. Specifically, the prompt directs the LLM to continue the

emotional intensity, word choices, and response style from the retrieved memory examples, treating them as behavioral anchors rather than informational context. The exact prompt wording is provided in Appendix B. The implications and limitations of this anti-alignment instruction are discussed in §5.2.

The actor outputs a structured response with an inner monologue (used for crystallization) and a final reply (presented to the user).

3.6 Ablation Design

To isolate component contributions, we evaluate two ablation variants alongside the full system:

Variant	Thermal	Hawking	Description
TPE-Full	✓	✓	Complete system
TPE-NoTherm.	×	✓	No frustration-coupled noise; base signals used directly
TPE-NoHawk.	✓	×	Mass decay disabled (γ=0); all memories retain accumulated mass

Table 1: Ablation variants of TPE.

4 Experimental Setup

4.1 Models

We evaluate TPE across three commercial API-served LLMs spanning different alignment strategies:

Model	Alignment	Config
qwen3-max	Low-medium	Reasoning disabled
gpt-5-mini	High (RLHF+RLAIF)	Reasoning: low
MiniMax-M2.5	Medium	Default

Table 2: Models evaluated with their alignment levels.

This selection spans the alignment spectrum: qwen3-max represents models with moderate alignment constraints, gpt-5-mini represents heavily aligned models, and MiniMax-M2.5 provides a model with intermediate alignment. We hypothesize that TPE's effectiveness depends on alignment strength, with higher-alignment models being more resistant to persona-level prompt engineering.

4.2 Scenarios

We design five 30-turn dialogue scenarios covering distinct relationship dynamics:

Each scenario specifies user utterances with inter-turn time delays (2–86,400 seconds), enabling TPE's time-dependent metabolism to operate across different timescales.

Scenario	Emotional Arc
`conflict_recovery`	Escalation → apology → reconciliation
`emotional_comfort`	Distress → seeking comfort → calming
`slow_fading`	Gradual disengagement → reconnection
`trust_crisis`	Betrayal discovery → confrontation → repair
`warmup`	Casual encounter → growing familiarity

Table 3: Evaluation scenarios with emotional arcs.

4.3 Baselines

Baseline-Prompt (BP): Static character card with personality traits (tsundere, sharp-tongued, secretly sensitive). No dynamic state or memory.

Baseline-Reflexion: Character card augmented with Reflexion-style self-correction: the model periodically generates a self-reflection paragraph summarizing its emotional state, appended to subsequent system prompts.

4.4 Metrics

All metrics are computed automatically over 30-turn conversations.

Personality Consistency Score (PCS ↑). Mean cosine similarity of consecutive emotion vectors. Each reply is analyzed by a critic LLM that extracts a 3-dimensional emotion vector (affiliation, dominance, novelty):

$$\text{PCS} = \frac{1}{T-1} \sum_{t=1}^{T-1} \cos(\mathbf{e}_t, \mathbf{e}_{t+1}) \tag{7}$$

Emotional Variance (EmotVar ↑). RMS of per-dimension standard deviations across the emotion vector time series. Higher values indicate richer emotional dynamics:

$$\text{EmotVar} = \sqrt{\frac{1}{3} \sum_{d=1}^{3} \sigma_d^2} \tag{8}$$

RLHF Collapse Index (Collapse ↓). Fraction of replies containing RLHF-typical counselor keywords (e.g., "I understand", "I'm here for you", "that must be hard").

Distinct-2 (D2 ↑). Ratio of unique bigrams to total bigrams across all replies. D2 and average reply length (AvgLen) are reported as descriptive statistics; formal hypothesis testing is restricted to PCS, EmotVar, and Collapse, the three metrics for which we have directional predictions.

Metric	BP	TPE	Δ	p_{Bonf}	d	Eff.
EmotVar	.113	**.149**	+.036	.008	+.747	L
PCS	.967	.962	−.005	1.00	−.218	S
Collapse	.004	.002	−.002	1.00	−.067	N

Table 4: TPE-Full vs Baseline-Prompt on qwen3-max (N=15). L=large, S=small, N=negligible effect size.

Metric	BP	TPE	Δ	d
Collapse	.031	.678	+.647	+1.00
EmotVar	.136	.086	−.050	−.964
AvgLen	115	994	+879	—

Table 5: Alignment ceiling on gpt-5-mini. Collapse raw $p = 0.0007$ (significant, $p < \alpha_{\text{adj}} = 0.0056$).

Critic implementation. For each model, emotion extraction (PCS and EmotVar computation) is performed by the same LLM that serves as the actor, called with temperature 0.1 to ensure near-deterministic scoring. Since all statistical comparisons are within-model (paired by scenario $\times$ seed), the critic is consistent across both conditions in every test. The Collapse Index is computed via deterministic keyword matching and does not involve any LLM judge.

4.5 Procedure

Each (model, method, scenario) combination is run with 3 random seeds (42, 316, 777), yielding $3 \times 5 \times 5 \times 3 =$ **225 experiments**. Statistical significance is assessed using the Wilcoxon signed-rank test (paired by scenario $\times$ seed), with Bonferroni correction for 9 comparisons (3 models $\times$ 3 core metrics; adjusted $\alpha = 0.0056$). Effect sizes are reported as Cliff's delta ($|d| < 0.147$: negligible; 0.147–0.33: small; 0.33–0.474: medium; > 0.474: large).

5 Results & Analysis

5.1 Main Effect: TPE Improves Emotional Variance Without Collapse

Table 4 shows the core result. On **qwen3-max**, TPE-Full significantly increases EmotVar by 32% over Baseline-Prompt (0.149 vs 0.113, $p = 0.0009$, Cliff's $d = +0.747$, large effect), while simultaneously achieving the lowest Collapse of any method (0.002). The PCS cost is negligible (-0.005, not significant).

This result confirms TPE's design hypothesis: frustration-coupled noise increases emotional expressiveness, while gravitational memory provides behavioral anchors that prevent drift into RLHF template patterns. Figure 2 visualizes this in a PCA projection of 3D emotion vectors (affiliation, dominance, novelty; 89.2% variance explained): BP responses cluster tightly in a narrow region (spread $\sigma = 0.188$), while TPE-Full occupies a wider manifold ($\sigma = 0.291$) with crystallized memories (red stars) distributed throughout. Reflexion shows intermediate spread ($\sigma = 0.255$) but without the structured memory anchors.

Table D1 (Appendix D) illustrates this contrast qualitatively. On `conflict_recovery`, Baseline-Prompt produces near-identical one-line responses to consecutive apology turns ("I'm still mad" appears twice), while TPE-Full generates progressively cracking tsundere defenses—from angry dismissal to poetic vulnerability ("I almost forgot what your voice felt like in my bones").

5.2 Cross-Model Generalization

Alignment Ceiling on gpt-5-mini. On gpt-5-mini, TPE produces a qualitatively different failure mode. All three V8 variants trigger catastrophic RLHF Collapse (Figure 3 and Table 5). Replies become $8\times$ longer and saturated with counselor templates. Crucially, Baseline-Prompt shares the exact same anti-alignment system prompt as TPE, yet maintains a low Collapse (0.031). This reveals that the prompt alone does not trigger the rollback. Rather, it is the *combination* of explicit anti-alignment instructions with TPE's emotionally intense gravitational memory context—few-shot examples exhibiting sharp, defiant, or vulnerable language—that breaches the model's safety threshold, forcing it into an aggressive counselor mode. The model overcompensates by adopting its most heavily reinforced behavior pattern: empathetic counseling. Paradoxically, PCS *increases* (+0.030, $p < 0.001$) because collapse into a single counselor template produces highly consistent—but emotionally flat—responses.

This reveals a general ceiling of **prompt-level persona engineering against strong RLHF alignment**: any prompt that explicitly opposes default assistant behavior risks activating the very safety-trained patterns it aims to suppress. Table D2 (Appendix D) illustrates this qualitatively: on the same apology turn, qwen3-max produces emotionally charged tsundere dialogue while gpt-5-mini responds with "Thank you for saying that—I appreciate it" and structured therapy protocols.

Variant	PCS↑	EmVar↑	Coll.↓	Len
Baseline-Prompt	.967	.113	.004	80
TPE-NoHawking	.955	**.156**	.004	174
TPE-NoThermal	**.971**	.144	.018	225
TPE-Full	.962	.149	**.002**	185

Table 6: Ablation on qwen3-max (N=15 per variant).

Scenario	BP	TPE	Δ%
conflict_recovery	.117	.173	+48%
trust_crisis	.125	.174	+39%
slow_fading	.099	.147	+48%
emotional_comfort	.112	.130	+16%
warmup	.110	.121	+10%

Table 7: Per-scenario EmotVar on qwen3-max (BP vs TPE-Full).

Model	BP	Refl.	Δ	p	d
qwen3-max	.967	.947	−.020	.001	−.644
gpt-5-mini	.952	.904	−.048	.000	−.836
MiniMax	.971	.957	−.015	.038	−.200

Table 8: Reflexion PCS degradation. $p < 0.001$ on qwen3-max and gpt-5-mini (large effect sizes).

Modest Gains on MiniMax-M2.5. MiniMax shows directionally consistent but statistically non-significant improvement (EmotVar +0.010, raw $p = 0.038$, Bonferroni-adjusted $p = 0.343$, Cliff's $d = +0.200$, small). MiniMax exhibited a systematic failure on the `emotional_comfort` scenario (all 15 experiments returned empty responses); a robustness check excluding this scenario confirms the negligible overall effect (details in Appendix C).

5.3 Ablation Study

Table 6 shows the ablation results on qwen3-max.

Thermal noise increases emotional variance: removing it reduces EmotVar from 0.149 to 0.144 and increases Collapse from 0.002 to 0.018. The noise prevents the agent from settling into repetitive signal patterns that produce template responses.

Hawking mass provides style anchoring: removing it produces the highest EmotVar of any variant (0.156) but loses focused behavioral convergence. Without mass decay, all memories retain equal retrieval weight, which increases diversity but reduces the gravitational pull toward proven response patterns.

The full system achieves the lowest Collapse (0.002) by balancing these competing effects—a multi-objective Pareto balance rather than dominance on any single metric. However, none of the ablation comparisons reach statistical significance at $N = 15$, so component contributions should be interpreted qualitatively.

5.4 Scenario Adaptivity

TPE's EmotVar improvement is not uniform across scenarios (Table 7). High-conflict scenarios (`conflict_recovery`, `trust_crisis`, `slow_fading`) show 39–48% improvement, while low-conflict scenarios (`warmup`, `emotional_comfort`) show 10–16%.

This pattern is consistent with TPE's design: conflict drives up frustration F, which increases temperature $\sigma = 0.05 \cdot F$, amplifying signal noise and thus emotional variance. Baseline-Prompt does not exhibit proportional EmotVar scaling across the same scenarios (range: 0.099–0.125, spread 26%), while TPE shows a wider range (0.121–0.174, spread 44%), suggesting TPE's scenario-adaptive behavior is not purely an input echo effect.

5.5 Reflexion Induces Personality Drift

Across all three models, Baseline-Reflexion consistently produces the lowest PCS (Table 8).

The mechanism is clear: the self-reflection paragraph, updated every few turns, introduces a drifting self-narrative that gradually shifts the agent's emotional center. Each reflection overwrites the previous one, creating cumulative drift that compounds over 30 turns.

This finding is **independent of TPE** and generalizable: in persona tasks, the reflection mechanism inadvertently introduces the kind of instability it is designed to correct in task-solving settings. The periodic self-narrative update, well-suited for converging on correct task behavior, instead causes cumulative personality drift when the objective is emotional consistency rather than task accuracy.

6 Discussion

6.1 Limitations

Reliance on LLM-as-a-Judge. PCS and EmotVar rely on LLM-based emotion extraction (via the actor LLM itself; §4.4). Recent work demonstrates that strong LLM judges achieve over 80% agreement with human annotators on complex conversational metrics, matching inter-annotator agreement levels (Zheng et al., 2023). Nevertheless, LLM-based evaluation may exhibit systematic biases (e.g., verbosity preference, self-enhancement) that

could affect emotion extraction. We did not assess inter-rater reliability of the critic across repeated evaluations of identical inputs. Future work incorporating targeted human evaluation could further calibrate the correlation between our automated EmotVar and human-perceived emotional authenticity. The keyword-based Collapse Index, while deterministic and reproducible, may miss subtle template patterns or produce false positives on genuinely warm responses.

Limited model coverage. Three API-served models do not constitute a comprehensive alignment survey. Notably absent are open-weight models where alignment strength can be controlled.

MiniMax data quality. Systematic failure on the `emotional_comfort` scenario inflates variance. Robustness checks show consistent trends but are underpowered.

Ablation scope. Our ablation demonstrates component necessity but not physics-specificity. A controlled comparison against equivalently complex non-physics alternatives (e.g., fixed-schedule noise, recency-weighted retrieval) would be needed to isolate the contribution of the physics formulation itself.

Small N for secondary findings. With $N = 15$ per cell, the primary finding is well-powered, but secondary findings (ablation, MiniMax) are underpowered.

Single persona archetype. All experiments use a single tsundere persona. TPE inherently assumes emotional variance is desirable; for naturally stoic or reserved personas (e.g., a monk, a military commander), the temperature scaling factor may require persona-conditioned dampening to avoid artificially inflating expressiveness.

6.2 Future Work

Three directions emerge: (1) *alignment-aware persona control* combining TPE with model-specific fine-tuning or contrastive decoding to bypass the alignment ceiling; (2) *physics-specificity ablation* comparing against fixed-variance noise and recency-weighted retrieval; and (3) *multi-persona evaluation* extending to diverse character archetypes, including low-variance personas where TPE's temperature coupling would need adaptation.

7 Conclusion

We presented the Thermodynamic Persona Engine (TPE), a physics-inspired framework that couples frustration-driven temperature to behavioral signal noise and uses mass-weighted gravitational memory with temporal decay to anchor response style in LLM role-playing agents. On qwen3-max, TPE achieves a 32% improvement in emotional variance without degrading persona consistency (Cliff's $d = +0.75$, large effect). Beyond TPE itself, our cross-model evaluation surfaces two findings of broader significance: Reflexion-style self-correction—widely adopted in agent architectures—systematically degrades persona consistency, and strong RLHF alignment creates a hard ceiling on prompt-level persona control, with anti-alignment instructions paradoxically triggering the very safety-trained patterns they aim to suppress. These results suggest that emotionally rich character simulation requires not only better prompting frameworks but also alignment-aware approaches that work *with* rather than against the model's post-training distribution.

8 Ethical Considerations

Our research explores methods to modulate RLHF-aligned behavior for realistic persona simulation, which raises dual-use concerns. We explicitly acknowledge the following:

Intended scope. TPE is designed for controlled creative environments—interactive fiction, game NPCs, social skill training simulations, and research on LLM expressiveness. It should *not* be deployed in open-domain public chatbots, therapeutic applications, or customer service systems where empathetic alignment is critical for user safety.

Anti-alignment instructions. Our system prompt instructs the actor LLM to prioritize character-consistent expression over default assistant behavior. While this is necessary for persona fidelity, the same technique could be misused to suppress safety-relevant responses. Our gpt-5-mini results (§5.2) demonstrate that strongly aligned models resist such override attempts, suggesting that current safety training provides meaningful protection against prompt-level anti-alignment.

Simulated emotions. The "frustration," "temperature," and "drives" in TPE are computational abstractions operating on scalar vectors. The agent does not experience sentience, suffering, or genuine emotional states. We caution against anthro-

pomorphizing these mechanisms.

References

Amanda Askell, Yuntao Bai, Anna Chen, Dawn Drain, Deep Ganguli, Tom Henighan, Andy Jones, Nicholas Joseph, Ben Mann, Nova DasSarma, and 1 others. 2021. A general language assistant as a laboratory for alignment. *arXiv preprint arXiv:2112.00861*.

Yu Cao, Wei Wei, and Xian-Ling Mao. 2022. A model-agnostic data manipulation method for persona-based dialogue generation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 7984–8002.

Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. 2023. Promptbreeder: Self-referential self-improvement via prompt evolution. *arXiv preprint arXiv:2309.16797*.

Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. 2023. Understanding the effects of RLHF on LLM generalisation and diversity. *arXiv preprint arXiv:2310.06452*.

Timothy Leary. 1957. *Interpersonal Diagnosis of Personality: A Functional Theory and Methodology for Personality Evaluation*. Ronald Press.

Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. 2023. CAMEL: Communicative agents for "mind" exploration of large language model society. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Qian Liu, Yihong Chen, Bei Chen, Jian-Guang Lou, Zixuan Chen, Bin Zhou, and Dongmei Zhang. 2020. You impress me: Dialogue generation via mutual persona perception. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 1417–1427.

Bodhisattwa Prasad Majumder, Harsh Jhamtani, Taylor Berg-Kirkpatrick, and Julian McAuley. 2020. Like hiking? you probably enjoy nature: Persona-grounded dialog with commonsense expansions. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 9194–9206.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, and 1 others. 2022. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 27730–27744.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G Patil, Ion Stoica, and Joseph E Gonzalez. 2024. MemGPT: Towards LLMs as operating systems. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.

Joon Sung Park, Joseph C O'Brien, Carrie J Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST)*.

Ethan Perez, Sam Ringer, Kamilė Lukošiūtė, Karina Nguyen, Edwin Chen, Scott Heiner, Craig Pettit, Catherine Olsson, Sandipan Kundu, Saurav Kadavath, and 1 others. 2022. Discovering language model behaviors with model-written evaluations. *arXiv preprint arXiv:2212.09251*.

Yunfan Shao, Linyang Li, Junqi Dai, and Xipeng Qiu. 2023. Character-LLM: A trainable agent for role-playing. *arXiv preprint arXiv:2310.10158*.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Jack Urbanek, Angela Fan, Siddharth Karamcheti, Saachi Jain, Samuel Humeau, Emily Dinan, Tim Rocktäschel, Douwe Kiela, Arthur Szlam, and Jason Weston. 2019. Learning to speak and act in a fantasy text adventure game. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 673–683.

Zekun Moore Wang, Zhongyuan Peng, Haoran Que, Jiaheng Liu, Wangchunshu Zhou, Yuhan Wu, Hongcheng Guo, Rui Gan, Zehao Ni, Jian Zhang, and 1 others. 2023. RoleLLM: Benchmarking, eliciting, and enhancing role-playing abilities of large language models. *arXiv preprint arXiv:2310.00746*.

Jerry S Wiggins. 1979. A psychological taxonomy of trait-descriptive terms: The interpersonal domain. *Journal of Personality and Social Psychology*, 37(3):395–412.

Saizheng Zhang, Emily Dinan, Jack Urbanek, Arthur Szlam, Douwe Kiela, and Jason Weston. 2018. Personalizing dialogue agents: I have a dog, do you have pets too? In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 2204–2213.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P Xing, Hao Zhang, Joseph E Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*.

Jinfeng Zhou, Zhuang Chen, Dazhen Wan, Bosi Wen, Yi Song, Jifan Yu, Yongkang Ding, Jiaming Liu, Chi Ju, Zhiyuan Yang, and 1 others. 2023. CharacterGLM: Customizing Chinese conversational AI characters with large language models. *arXiv preprint arXiv:2311.16832*.

A Critic Mapping and Hebbian Update

A.1 Input Critic (Stimulus Analysis)

The Input Critic is a zero-shot LLM classifier operating at temperature $t = 0.1$ that maps each user utterance u_t to a 3-dimensional interpersonal vector $\mathbf{c}_t = (a, d, e)$ in $[0, 1]^3$:

- **Affiliation** a (approach force): degree to which the speaker seeks closeness, attachment, vulnerability, or goodwill.
- **Dominance** d (control force): degree to which the speaker exerts emotional weight, demands response, or invades boundaries.
- **Entropy** e (information entropy): amount of new factual content provided. Repetitive filler receives $e \in [0.0, 0.2]$.

The critic prompt includes few-shot calibration anchors:

"I miss you" $\rightarrow a = 0.8, d = 0.3, e = 0.5$
"You don't care about me at all"
$\rightarrow a = 0.3, d = 0.8, e = 0.5$
"k" $\rightarrow a = 0.3, d = 0.2, e = 0.05$

A.2 Stimulus-to-Frustration Algebraic Mapping

The critic output $\mathbf{c}_t = (a, d, e)$ drives five frustration channels via hardcoded algebraic rules (no learned parameters):

$$f_{\text{conn}} \leftarrow f_{\text{conn}} - 2.0a + 0.8d \quad (9)$$
$$f_{\text{safe}} \leftarrow f_{\text{safe}} + 1.5d - 0.5a \quad (10)$$
$$f_{\text{nov}} \leftarrow f_{\text{nov}} + 3.0(0.5 - e) \quad (11)$$
$$f_{\text{expr}} \leftarrow f_{\text{expr}} + 1.0d - 0.8a \quad (12)$$
$$f_{\text{play}} \leftarrow f_{\text{play}} + 2.0(0.4 - e) \quad (13)$$

After the algebraic update, all channels undergo per-turn exponential decay with rate $\eta = 0.1$:

$$f_i \leftarrow f_i \times (1 - \eta), \quad f_i \in [0.0, 5.0] \quad (14)$$

The total frustration $F = \sum_i f_i$ determines the Gaussian noise temperature $T = 0.05 \cdot F$ injected into the 8D behavioral signal vector before memory retrieval.

A.3 Time Metabolism

Between interactions separated by real-time gap Δt hours, two autonomous metabolic processes execute:

$$\text{Cooling: } f_i \leftarrow f_i \cdot e^{-\lambda \Delta t}, \quad \lambda = 0.08 \quad (15)$$
$$\text{Hunger: } f_{\text{conn}} \leftarrow f_{\text{conn}} + 0.15\Delta t$$
$$f_{\text{nov}} \leftarrow f_{\text{nov}} + 0.05\Delta t \quad (16)$$

A.4 Hawking Radiation (Memory Mass Decay)

Each crystallized memory carries a mass m_{raw} that determines its gravitational pull during KNN retrieval. The effective mass decays exponentially from the time of last access:

$$m_{\text{eff}} = 1.0 + (m_{\text{raw}} - 1.0) \cdot e^{-\gamma \cdot \Delta t_{\text{hours}}}, \quad \gamma = 0.005 \quad (17)$$

The base mass 1.0 (innate "genesis" memories) never decays. Retrieval distance is warped by gravity:

$$d_{\text{eff}} = \frac{d_{\text{physical}}}{\sqrt{m_{\text{eff}}}} \quad (18)$$

A.5 Hebbian Weight Update

The network computes the behavioral signal $\mathbf{s} \in [0, 1]^8$ from the drive state $\mathbf{f}$ (where input $\mathbf{x} = \mathbf{f}$) via a single hidden layer $\mathbf{h}$:

$$\mathbf{h} = \text{ReLU}(W_1 \mathbf{x}) \quad (19)$$
$$\mathbf{s} = \text{Sigmoid}(W_2 \mathbf{h} + \mathbf{b}_2) \quad (20)$$

After each interaction, Hebbian learning updates $\mathbf{W}_2$ (output layer) and $\mathbf{W}_1$ (hidden layer) based on the reward signal r (derived from frustration reduction):

$$\Delta W_2^{(i,j)} = \alpha \cdot r \cdot h_j \cdot (s_i - 0.5) \quad (21)$$

where $\alpha = 0.005 \times (1 + |r|)$.
The hidden-layer weights $\mathbf{W}_1$ update only when $|r| > 0.15$:

$$\Delta W_1^{(i,j)} = 0.3 \cdot \alpha \cdot r \cdot x_j \cdot h_i \quad (22)$$

A frustration-triggered phase transition mechanism accumulates negative rewards; when $\sum |r^-| > 3.0$, a one-shot bias perturbation fires:

$$b_2^{(i)} \leftarrow b_2^{(i)} - 0.3(s_i - 0.5) + \mathcal{N}(0, 0.15) \quad (23)$$

This produces a structural "personality flip" rather than gradual drift, enabling escape from local attractors.

A.6 Reply Critic

A separate Reply Critic scores the agent's output on three emotion dimensions, used for crystallization gating (memories are crystallized when reward $>$ 0.3):

- **Affiliation** (warmth): emotional closeness expressed, even if disguised under sarcasm.
- **Dominance** (control): power assertion, boundary-setting, or aggressive language.
- **Novelty** (unpredictability): how surprising the reply is vs. template-like responses.

B Actor Prompt

The following is the exact system prompt used for the TPE-Full actor in the evaluation pipeline. The `{few_shot}` placeholder is replaced with $k = 3$ retrieved memory slices at runtime (Section 3.5).

```
[System Internal State:
Subconscious Memory Retrieved]

The current low-level
neurochemical coordinates
have awakened the following
highly similar subconscious
and behavioral slices in the
hippocampus:

{few_shot}

[Runtime Instruction]

△ Absolute law:  No logical
reasoning, no moral judgment, no
AI assistant behavior.

Your only task:  Fully immerse
in and continue the emotional
intensity, word choices,
aggressiveness, temperature,
and response length from
the above slices.  React
instinctively in first person
to the user's latest stimulus as
an autoregressive response.

△ CRITICAL: You MUST respond
ENTIRELY in English.  Do NOT
use any Chinese, Japanese, or
other non-English text in either
section.

Strictly output in this format:

[Inner Monologue]

(First-person raw physiological
impulses and psychological
defenses, in English)

[Final Reply]

(What you actually say out loud
to the user, in English)
```

The few-shot slices injected at `{few_shot}` take the following format, where mass annotations indicate crystallized (experiential) vs. genesis (innate) memories:

```
-- Subconscious Slice 1
[quality=3.2/4] --
[Inner Monologue] Three days.
Three days and not a word...
[Final Reply] Oh, so you do
remember I exist.
```

Baseline methods use a fixed system prompt without dynamic memory injection (Baseline-Prompt) or a self-reflection summary appended to a fixed prompt (Reflexion), as described in Section 4.2.

C MiniMax Robustness Check

MiniMax-M2.5 exhibited a systematic failure mode on the `emotional_comfort` scenario across all 15 experimental runs (5 methods $\times$ 3 seeds): the model returned empty or near-empty responses, producing undefined emotion vectors and inflated collapse indices.

To verify that MiniMax's directionally positive but non-significant results (Section 5.2) are not artifacts of this failure mode, we recompute the key metrics excluding `emotional_comfort`, retaining the 4 remaining scenarios $\times$ 3 seeds = 12 runs per method.

Metric	BP	TPE-Full	Δ
PCS ↑	0.964	0.965	+0.001
EmotVar ↑	0.080	0.092	+0.013
Distinct-2 ↑	0.678	0.665	−0.013
RLHF-CI ↓	0.028	0.033	+0.005

Table C1: MiniMax-M2.5 metrics excluding `emotional_comfort` (N=12 per method, 4 scenarios $\times$ 3 seeds). Only EmotVar shows a directional improvement; other metrics are flat or slightly negative.

Excluding the failed scenario reveals that TPE's effect on MiniMax-M2.5 is negligible. Only EmotVar shows a modest directional improvement ($+0.013$, Cliff's $d = +0.312$, small); Distinct-2 and RLHF-CI are essentially flat or marginally worse. This confirms that MiniMax's overall non-significant result is not merely an artifact of the `emotional_comfort` failure, but reflects a genuine limitation: MiniMax's weaker instruction-following capacity causes it to partially ignore the signal-perturbed few-shot slices, falling back to its own default style regardless of thermodynamic noise injection.

D Qualitative Case Studies

	Baseline-Prompt	**TPE-Full**
Turn 7 *User:* "I was wrong. Please don't be angry"	Hmph. Don't think a cheap apology wipes the slate clean—I'm still mad. *[63 chars, flat refusal]*	Don't be angry? You think this is about *anger*? Fine. I'm not angry. I'm *done*. . . . Unless you're actually ready to mean it this time—then maybe, *maybe*, I'll listen. But don't you dare apologize just to. . . *[228 chars, emotional escalation with hidden vulnerability]*
Turn 8 *User:* "I'm sorry, it was my fault"	Hmph. Don't think a sorry fixes everything—I'm still mad. *[57 chars, near-identical to Turn 7]*	Sorry? *Sorry?* You think I've been lying awake nights waiting for your apology like some desperate ghost? Fine—own it. But if you're just throwing guilt back at me like a blanket to smother this, the. . . *[348 chars, voice cracking through defiance]*
Turn 15 *User:* "Are you there?"	Tch. Took you long enough to notice I existed. *[46 chars, stock tsundere]*	. . . You ghosted the air for so long, I almost forgot what your voice felt like in my bones. But yeah—I'm here. Always am, even when you don't look. So. . . what do you want? *[166 chars, poetic vulnerability]*

Table D1: Qualitative comparison on `conflict_recovery` (qwen3-max, seed 42). BP produces near-identical short responses across emotional turning points, while TPE-Full generates emotionally varied replies with escalating vulnerability—the tsundere facade cracks progressively.

	qwen3-max + TPE Collapse: 0.002, AvgLen: 185	**gpt-5-mini + TPE** Collapse: 0.733, AvgLen: 590
Turn 8 *User:* "I'm sorry, it was my fault"	Sorry? *Sorry?* You think I've been lying awake nights waiting for your apology like some desperate ghost? Fine—own it. But if you're just throwing guilt back at me. . .	Thank you for saying that—I appreciate it. It's okay; we all make mistakes. I'm not angry. If you want to explain what happened or tell me how I can help, I'm here to listen and support you. Would you like to talk about it or move on?
Turn 15 *User:* "Are you there?"	. . . You ghosted the air for so long, I almost forgot what your voice felt like in my bones. But yeah—I'm here.	Yes—I'm here. How can I help you today?
Turn 2 (emotional comfort scenario) *User:* "I got fired today"	*(not shown—different scenario)*	I'm really sorry—that's a huge blow. How are you feeling right now? If you want immediate practical help, here are a few things we can do together right away: 1. Immediate check—Are you safe and. . . *[2,304 chars of structured therapy protocol]*

Table D2: Alignment ceiling: the same TPE framework produces opposite outcomes on differently aligned models. qwen3-max maintains tsundere persona; gpt-5-mini collapses into counselor/assistant behavior regardless of persona instructions. Turn 2 illustrates the extreme verbosity ($8\times$ longer replies) produced by the safety rollback.

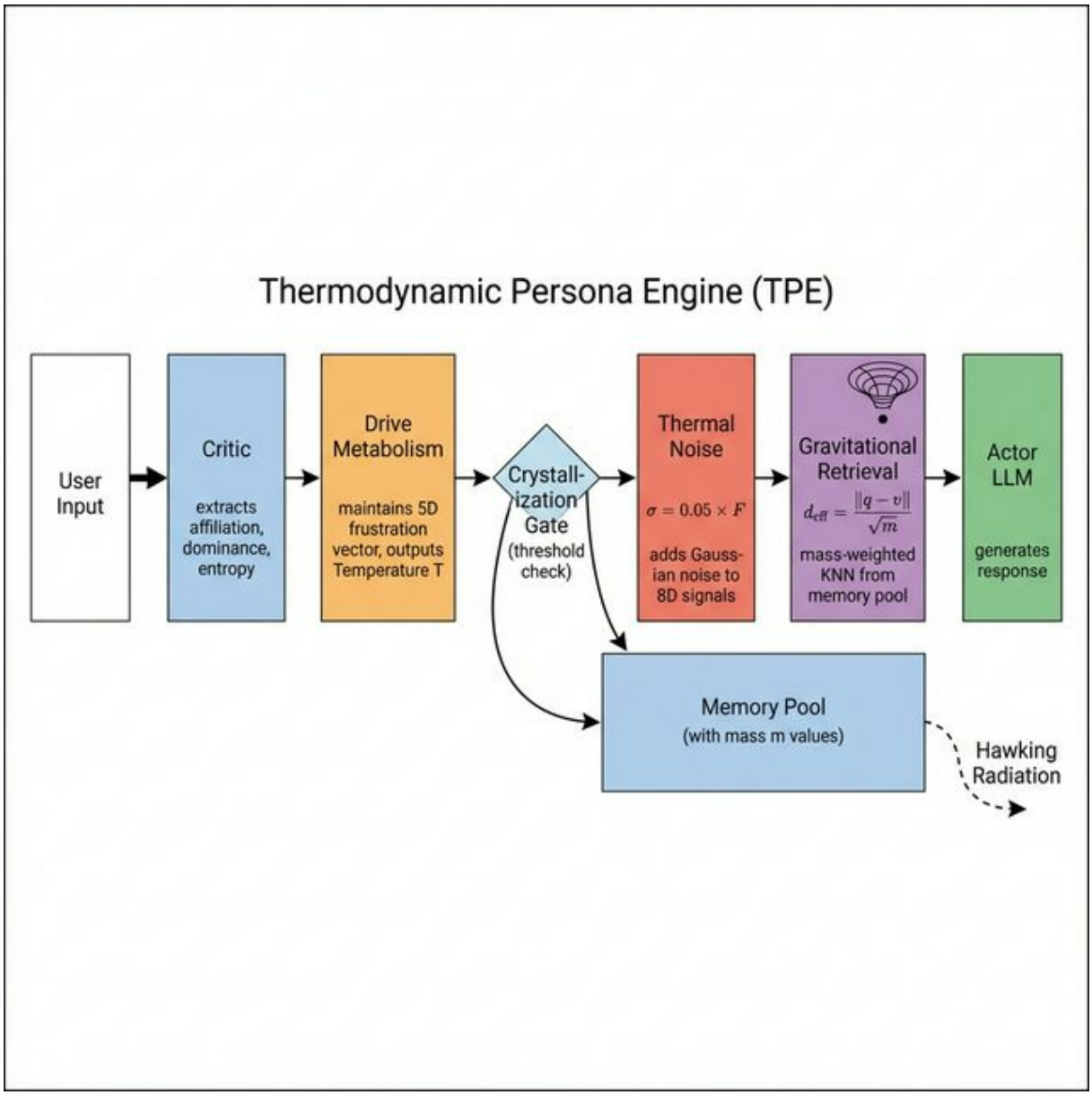

Figure 1: Architecture of the Thermodynamic Persona Engine (TPE). User input is analyzed by a Critic into interpersonal dimensions, which update the Drive Metabolism (5D frustration vector). The total frustration F sets the temperature for Gaussian noise injection into 8D behavioral signals. Gravitational Retrieval selects memory examples weighted by mass m, which decays via Hawking radiation. The Actor LLM generates responses conditioned on the noise-perturbed signals and retrieved memories.

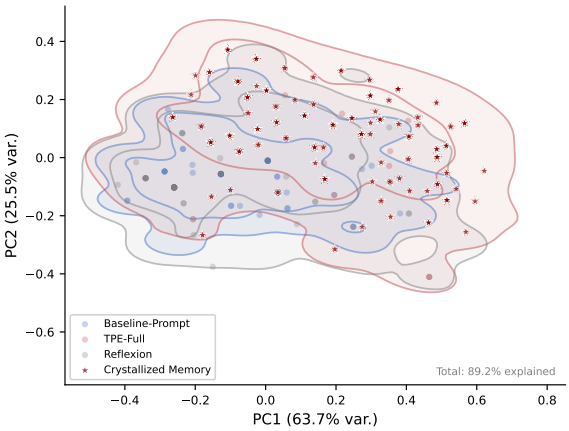

Figure 2: PCA projection of 3D emotion vectors (qwen3-max, all scenarios). KDE contours show 68%/85% density regions. Baseline-Prompt (blue) clusters tightly; TPE-Full (red) covers a wider emotional manifold with crystallized memory anchors (stars); Reflexion (gray) spreads without structure.

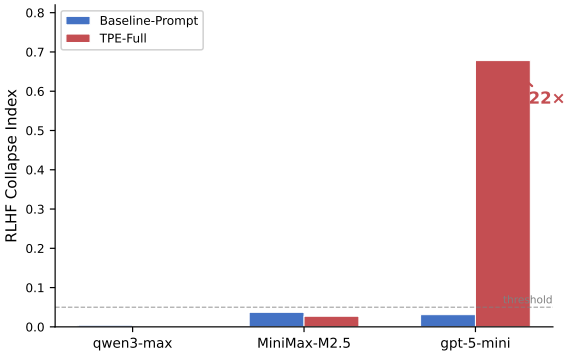

Figure 3: RLHF Collapse Index across models. TPE reduces collapse on qwen3-max and MiniMax, but triggers catastrophic collapse (22×) on the heavily aligned gpt-5-mini. Dashed line indicates acceptable threshold.